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Research Project Report on the Topic:
A/B Testing of a Recommender System as a Black-Box Experiment

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Abstract

This project studies black-box optimisation of user-interface configurations under a simulated A/B testing environment. A web call-to-action button is parameterised by twelve continuous variables (background and text colour, geometric dimensions, font size, text quality, padding, vertical position, hour of day). The click-through rate (CTR) as a function of these parameters is treated as an expensive noisy black-box function: every evaluation simulates an A/B test with a finite user sample and Bernoulli outcomes. Three derivative-free optimisers are compared — DIRECT, Differential Evolution (DE), and Bayesian Optimisation with Gaussian processes (BO + EI) — under two distinct simulator realisations: a ground-truth analytical formula derived from empirical UX research (WCAG 2.1, Baymard 2024, Chartbeat 2014, Nielsen Norman Group 2018, Itten 1961, Ou & Luo 2006), and a trained logistic regression surrogate fitted to 120,000 synthetic samples drawn from the formula. Cross-validation of ML-found optima against the oracle reveals that BO systematically overestimates the true CTR by a factor of roughly $3.5\text{--}4\times$ (gap of order $+0.4\text{--}+0.5$ across five independent seeds), while DIRECT and DE remain within approximately $+0.1$ of the truth. The exact magnitudes vary across runs because the synthetic dataset, the trained surrogate, and the optimisers themselves are re-randomised on each invocation of the pipeline, but the qualitative ordering of the three methods reproduces consistently. On the honest simulator BO and DIRECT are statistically indistinguishable (0.335 ± 0.011 vs 0.327 ± 0.019), yet BO requires runtime roughly 7000 times larger. The principal finding is that surrogate-based black-box optimisation can produce overoptimistic conclusions that cannot be detected in production without out-of-distribution validation, making DIRECT the preferred choice by the joint criterion of quality, stability, and computational cost.

Аннотация

Проект посвящён сравнению методов оптимизации чёрного ящика на задаче поиска конфигурации веб-кнопки, максимизирующей CTR. Кнопка описывается двенадцатью непрерывными параметрами (цвет фона и текста, размер, шрифт, качество текста, отступы, позиция на странице, время суток). CTR как функция параметров рассматривается как дорогая шумная чёрная функция: каждый вызов имитирует A/B-тест на конечной выборке пользователей с бернуллиевскими событиями. Сравниваются три безградиентных метода --- DIRECT, дифференциальная эволюция (DE) и байесовская оптимизация с гауссовыми процессами (BO + EI) --- при двух реализациях симулятора: аналитическая формула на основе эмпирических UX-исследований и обученная логистическая регрессия (суррогат). Cross-validation оптимумов, найденных через суррогат, относительно оракула показал, что BO систематически переоценивает истинный CTR примерно в $3,5\text{--}4$

раза (gap порядка $+0,4$ -- $+0,5$ по пяти прогонам), тогда как DIRECT и DE удерживают расхождение в районе $+0,1$. Конкретные численные значения варьируются от запуска к запуску, поскольку датасет, суррогат и сами оптимизаторы каждый раз инициализируются новым сидом, однако качественный порядок методов воспроизводится во всех прогонах. На честном симуляторе BO и DIRECT статистически неразличимы ($0,335 \pm 0,011$ против $0,327 \pm 0,019$), при этом BO требует в 7000 раз больше времени. Основной вывод: оптимизация поверх обученной модели может давать оверфит-результаты, которые невозможно отличить от истинных в продакшене, что делает DIRECT предпочтительным выбором по совокупности качества, устойчивости и вычислительной стоимости.

Keywords

black-box optimisation; A/B testing; DIRECT; differential evolution; Bayesian optimisation; Gaussian process; CTR; surrogate model; out-of-distribution extrapolation; web UI configuration.

Source Code

All code, datasets, unit tests, and pipeline scripts produced in the course of this work are publicly available in the project repository: https://github.com/StefanieTheSentinels/Course_project. The repository contains a reproducible end-to-end pipeline that regenerates every numerical result reported below from five different random seeds.

1 Introduction

1.1 Subject Area and Motivation

Modern digital products continuously tune the visual and behavioural configuration of user-facing elements — button colour, size, copy, placement, font weight, surrounding whitespace — to maximise downstream business metrics such as click-through rate (CTR), conversion, and long-run user retention. The standard industrial procedure is to compare two or more candidate configurations against a control variant by means of a live A/B test, exposing a controlled fraction of users to the new variant and measuring the observable outcome over the course of several days or weeks [18]. The cost of one such evaluation is far from negligible: statistical power against realistic effect sizes typically requires tens of thousands of users per variant, and meaningful secondary metrics often need even longer observation windows. The result is moreover intrinsically noisy, since user clicks are Bernoulli draws whose variance is bounded only by sample size.

Mathematically this is the canonical setting of black-box optimisation of a noisy expensive function: the optimiser may query the response function only through its scalar output, without access to its gradient or to any internal structure of the system that produces the output [2, 8]. The problem class admits no generic closed-form solution and is conventionally approached by derivative-free methods such as DIRECT [16], Differential Evolution (DE) [26], and Bayesian Optimisation (BO) with Gaussian process surrogates [24, 25]. Each of these methods trades exploration breadth, sample efficiency, and per-iteration computational overhead differently, and the choice between them is rarely obvious in advance.

A practical complication that distinguishes the industrial case from the textbook formulation is that the response surface itself is rarely available in closed form. Most product teams have access only to event logs and to a predictive model trained on those logs. It is tempting in that situation to use the predictive model as a cheap stand-in for the true response function, and to optimise against the surrogate rather than against live traffic. This procedure, however, exposes the optimiser to a hazard widely studied in offline reinforcement learning and policy evaluation but rarely measured explicitly in classical black-box optimisation literature: *out-of-distribution exploitation* [9]. The optimiser, seeking to maximise its objective, naturally drifts towards regions of parameter space where the surrogate predicts high reward; if those regions happen to lie far from the training support of the surrogate, the prediction may be arbitrarily wrong while still being arbitrarily high. The optimiser is in effect rewarded for fooling its own model.

The present work investigates the magnitude of this hazard on a realistic but fully controlled toy problem.

1.2 Problem Statement

Let $\theta \in \mathcal{X} \subset \mathbb{R}^{12}$ denote a vector of twelve UI parameters and let $p(\theta)$ denote the true click probability. Given an evaluation budget B of A/B-test invocations, find

$$\theta^* \in \arg \max_{\theta \in \mathcal{X}} \text{CTR}(\theta), \quad \text{CTR}(\theta) = \frac{1}{n} \sum_{i=1}^n \text{Bernoulli}(p(\theta)), \quad (1)$$

using only evaluations of the form $\theta \mapsto \text{CTR}(\theta)$. The optimiser must not access $p(\theta)$ directly, the gradient $\nabla p(\theta)$, or any internal state of the simulator.

1.3 Project Outcomes

The work delivers (i) a synthetic but empirically anchored ground-truth model of $p(\theta)$ based on published UX research; (ii) a reproducible black-box simulator that wraps any response model and emits noisy CTR estimates via Bernoulli sampling; (iii) an experimental comparison of DIRECT, DE, and BO on this simulator under both a formula-driven and a learning-based response model; (iv) a quantitative analysis of the extrapolation gap between surrogate-claimed CTR and oracle-true CTR; (v) open-source Python code with unit tests and a reproducible pipeline script [23].

2 Literature Review

The project draws on three lines of work that are usually read in isolation: derivative-free black-box optimisation, online controlled experimentation, and empirical web-usability research. The first provides the optimisation algorithms compared in this report; the second supplies the conceptual framing of the objective function as a controlled A/B test; the third supplies the empirical content of the ground-truth response surface against which all algorithms are benchmarked. Each individual source is summarised below, with explicit identification of how it is used in the present work.

2.1 Derivative-Free Optimisation

Jones, Perttunen and Stuckman (1993) introduced the DIRECT algorithm [16], a deterministic partitioning method for global optimisation that hierarchically divides the feasible hyperrectangle into smaller subrectangles, selecting at each iteration those subregions identified as “potentially optimal” according to a parametric trade-off between estimated function value and remaining region size. Their central contribution is a convergence proof that does not require knowledge of the Lipschitz constant of the objective, which makes DIRECT applicable to objectives that are merely continuous or even mildly discontinuous. In this project DIRECT serves as the canonical deterministic baseline against which stochastic and model-based methods are compared.

Storn and Price (1997) proposed Differential Evolution [26], a population-based heuristic in which a fixed pool of candidate solutions evolves through mutation, crossover, and tournament selection. The mutation step is the algorithm’s distinctive feature: each trial vector is constructed from a linear combination of three other randomly chosen population members, producing an exploration step whose magnitude self-adjusts to the current population dispersion. DE has become a default choice for problems where the objective is non-smooth or multimodal but evaluation is cheap. In this project DE represents the stochastic, gradient-free family of optimisers; its main weakness in the present setting is that the required number of evaluations grows with the product of population size and number of generations, which strains the limited evaluation budget.

Jones, Schonlau and Welch (1998) introduced the foundational framework of Efficient Global Optimization [17], which establishes Bayesian optimisation with Gaussian process surrogates and the Expected Improvement acquisition function as a principled response to the expensive black-box problem. Their formulation treats the objective as a sample path of a stochastic process and selects new query points by maximising the expected gain over the current best observation. The paper’s central insight — that an explicit probabilistic model of the objective enables provably sample-efficient exploration — underlies the entire BO branch of the field, and is the conceptual core of the BO method evaluated in this project.

Snoek, Larochelle and Adams (2012) adapted Bayesian optimisation to the practical needs of machine-learning hyperparameter tuning [25], addressing kernel choice, prior calibration, and parallel evaluation. Their empirical results established BO as the default sample-efficient choice for problems with moderately expensive evaluations and dimensionality up to roughly twenty. The skopt implementation used in this work follows their general framework, in particular the Matérn-5/2 kernel and the Expected Improvement acquisition.

Shahriari, Swersky, Wang, Adams and de Freitas (2016) provide the standard survey of Bayesian optimisation [24], classifying acquisition functions, surrogate models, and the design choices that distinguish modern BO from earlier response-surface methods. The survey explicitly discusses the trade-off between sample efficiency and per-iteration cost, which is the central argument exploited in the present experimental comparison: BO is sample-efficient but each step incurs $O(n^3)$ cost from the GP fit, which limits its competitive advantage when evaluations are not extremely expensive.

Frazier (2018) provides a tutorial-level introduction to Bayesian optimisation [8], with a clear derivation of the Expected Improvement acquisition function and a discussion of when BO is and is not the appropriate tool. The tutorial’s framing of BO as a strategy that “buys” sample efficiency at the price of modelling effort is used in this report as the conceptual lens through which the BO-versus-DIRECT trade-off is interpreted.

Audet and Hare (2017) present a comprehensive textbook treatment of derivative-free and black-box optimisation [2], covering direct-search methods, model-based methods, and stochastic methods within a unified framework. The book is the principal reference for the present project’s positioning of DIRECT, DE, and BO as three points in a common design space rather than as competing alternatives.

Eriksson, Pearce, Gardner, Turner and Poloczek (2019) introduced trust-region Bayesian optimisation [5], an extension of standard BO that restricts the acquisition function to a trust region around well-sampled points, thereby limiting the optimiser’s ability to exploit extrapolation regions of the surrogate. Although TuRBO is not implemented in this project, the present results give an experimental illustration of precisely the failure mode that motivated their work, and we identify it as the most natural next-step optimiser to evaluate.

2.2 A/B Testing and Online Experimentation

Rubin (1999) introduced the potential outcomes framework [22] that underlies the causal interpretation of randomised experiments. The framework formalises the counterfactual outcomes of treatment and control conditions for each unit and proves that random assignment is sufficient to identify the average treatment effect from observed data. This is the theoretical justification for treating an A/B test as a

measurement of causal effect, which in turn is what licenses the optimisation problem in equation (1) to be interpreted as a search for the configuration with the highest true click probability rather than merely the highest observed click rate.

Kohavi and Longbotham (2016) document the practical methodology of A/B testing in large-scale online systems [19], covering variant definition, the choice of randomisation unit, metric selection, and the most common operational pitfalls encountered in production. Their treatment is the source of the conceptual framework adopted in this project for what a “call” to the black box represents: a fully randomised exposure of a fixed user sample to a candidate configuration, with observable outcomes aggregated into a CTR estimate.

Kohavi, Tang and Xu (2020) extend the earlier methodological work into a comprehensive industrial guide to trustworthy online controlled experiments [18]. Their textbook codifies the modern industrial practice, including the role of guardrail metrics, the costs and frequency of running under-powered tests, and the treatment of secondary outcomes. The present work uses this reference as the conceptual source for treating CTR as the primary metric and the black-box simulator as an abstraction of a fully controlled experimental infrastructure.

Deng, Lu and Litz (2017) document several common analytical pitfalls in large-scale online A/B testing [4], with particular attention to the consequences of mismatches between the unit of randomisation and the unit of analysis, and to violations of the i.i.d. assumption in user-level data. The paper informs the project’s choice to model each black-box call as a sample of independent Bernoulli outcomes from a fixed click probability, which is the appropriate idealisation when randomisation and analysis are aligned at the user level.

Fabijan, Dmitriev, Gupchup, Omhover and Vermeer (2019) formalise Sample Ratio Mismatch (SRM) as a critical data-quality check in online experiments [6]. SRM occurs when the observed traffic split between control and treatment significantly deviates from the intended split, indicating a bug in randomisation, logging, or triggering. The paper provides detection procedures and a taxonomy of root causes. The present project implements a chi-square SRM check in its analysis module as standard practice, even though by construction the synthetic simulator cannot exhibit SRM in expectation.

Johari, Koomen, Pekelis and Walsh (2017) expose a subtle inferential pitfall of continuous A/B test monitoring [15]: when standard fixed-sample hypothesis tests are applied repeatedly as data accumulates, the effective Type I error rate inflates well above the nominal level. The phenomenon, known as “peeking,” invalidates the most common informal practice of watching a running experiment and stopping as soon as significance is achieved. The paper motivates the project’s choice to fix the number of black-box calls in advance rather than to stop adaptively.

Johari, Koomen and Pekelis (2022) develop the “always-valid inference” framework as a prin-

cipled response to the peeking problem [14], providing p -values and confidence intervals that remain valid under arbitrary continuous monitoring. The framework establishes the theoretical foundation for sequentially valid online experimentation. The present project does not implement sequential testing, but the framework is the natural reference for any future extension that lifts the fixed-budget assumption used in the experimental protocol.

2.3 Empirical UX Research Anchoring the Response Surface

The W3C Web Content Accessibility Guidelines 2.1 [28] specify a minimum contrast ratio of 4.5:1 for normal-size text against its background, derived from extensive psychophysical testing of contrast perception across the visually impaired population. The guideline is operationalised in the project’s ground-truth formula as a multiplicative penalty that saturates at this threshold: contrast below 4.5:1 attenuates click probability proportionally, while contrast above it provides no further benefit.

Baymard Institute (2024) reports a large-scale empirical audit of e-commerce interfaces [3] that establishes the 48×48 pixel touch target as the empirical minimum for reliable mobile interaction. Below this size, mis-tap rates rise sharply; above it the effect saturates. The threshold informs the size component of the ground-truth formula, which uses a saturating function that reaches unity at 48×48 on mobile and at 36×36 on desktop.

Apple Human Interface Guidelines [1] prescribe the same 44×44 pt minimum touch target (translating to approximately 48×48 px at standard density) and provide independent corroboration of the Baymard threshold. The convergence of guidance across the major design systems is what allows the threshold to be treated as an empirical regularity rather than a single platform convention.

Google Material Design 3 [10] reaches the same 48×48 dp recommendation and additionally emphasises the role of symmetric padding around interactive elements — a principle that informs the project’s margin-balance penalty, which attenuates click probability when the horizontal and vertical clearances inside a button are unequal.

Chartbeat’s 2014 analysis of two billion web sessions [11] established that user engagement with content drops sharply below the visible fold, with the engagement curve falling by roughly 60% per fold. The finding informs the steepness of the exponential decay used in the project’s scroll-position multiplier, calibrated so that scroll positions one fold below the optimum receive approximately $\exp(-3 \cdot 0.35) \approx 0.35$ of the maximum visibility.

Nielsen Norman Group (Fessenden, 2018) refines the above-the-fold conventional wisdom [7] by identifying that the primary call-to-action zone is 10–30% below the top of the visible page rather than at the very top, since the very top is typically occupied by navigation and value-proposition headers.

This finding motivates the sweet-spot model used in the project’s vertical-position function: a linear ramp from 0.85 at the page top to 1.00 at scroll position 0.15, followed by exponential decay.

Itten’s classical theory of colour [13] establishes the colour-wheel taxonomy of harmonic combinations (complementary, triadic, split-complementary, analogous, identical). The taxonomy provides the conceptual basis for the project’s colour-harmony multiplier, which assigns higher click probability to colour pairs that fall in established harmony classes and substantially lower probability to near-identical hues, where the text becomes effectively invisible against the background.

Ou and Luo (2006) provide an empirical validation of the Itten framework [21] through systematic experiments on two-colour combinations, fitting a quantitative model to observer preference ratings. Their results are the source of the specific numerical multipliers used in the project’s harmony function (0.05 for same-hue pairs, 0.70–1.00 for harmonic pairs depending on hue-angle separation).

Lidwell, Holden and Butler (2010) codify the principle of symmetric whitespace inside UI elements [20] as part of their broader catalogue of design principles. This principle is the source of the margin-balance multiplier in the project’s formula, which penalises buttons in which the horizontal clearance around the text substantially differs from the vertical clearance.

Infolinks (2014) reports CTR patterns across approximately one trillion ad impressions [12], identifying a diurnal peak around midday (11:00–14:00 local time) and a trough during the night. The pattern is operationalised in the project’s time-of-day multiplier with a 1.14 peak coefficient and a 0.90 night coefficient.

Visual Website Optimizer (VWO, 2014) documents a notable case study [27] in which a sign-up form on Open Mile’s landing page increased conversions by 232% after the surrounding whitespace was substantially increased. The case study motivates the sigmoid transition centred at whitespace ratio 0.2 used in the project’s formula — below this ratio the formula penalises crowding, above it the effect saturates.

2.4 Positioning of the Present Work

To the best of the author’s knowledge, no prior publicly available work compares DIRECT, DE, and BO on a synthetic-but-realistic CTR optimisation task while explicitly quantifying the extrapolation gap of a learned surrogate against an analytical oracle. The closest related approaches treat A/B testing solely as a hypothesis-testing problem [4] or focus on the surrogate quality without measuring downstream optimisation behaviour. The present work occupies the gap between those two literatures by treating A/B testing as an optimisation problem and asking how the choice of optimiser interacts with the choice of surrogate. The project is positioned as a small-scale, fully reproducible benchmark of that interaction.

3 Simulator Architecture and Ground-Truth Model

3.1 Black-Box Interface

The simulator is implemented as a Python class `BlackBox` whose sole public method accepts a parameter dictionary and returns a noisy CTR estimate (Listing 1). The optimisers in this work interact with this class through the single-line call `ctr = box(params).ctr` and have no access to the response model within. This is the technical realisation of the black-box postulate of equation (1). The class design is deliberately minimal: neither the response model nor the noise model is exposed in the public interface, and the optimiser is unable to distinguish at runtime whether it is interacting with the analytical oracle or with a trained ML surrogate.

Listing 1: Black-box interface used by all optimisers

```
1 def __call__(self, params, n_users=None):
2     p_click = self._get_p_click(params)
3     if self.noise_std > 0:
4         p_safe = np.clip(p_click, 1e-12, 1 - 1e-12)
5         logit = np.log(p_safe / (1 - p_safe))
6         logit += self.rng.normal(0, self.noise_std * 4)
7         p_click = 1.0 / (1.0 + np.exp(-logit))
8     clicks = self.rng.binomial(1, p_click, size=n_users)
9     return SimulationResult(ctr=clicks.mean(), ...)
```

Two response models are interchangeable inside the black box: a *ground-truth formula* (oracle) and a *trained ML model* (surrogate). The first allows controlled experiments where the global optimum is in principle attainable; the second emulates the realistic industrial scenario where only data, not formulas, are available. Noise is injected in logit space rather than in probability space, so that the variance of the perturbation does not collapse as $p \rightarrow 0$ or $p \rightarrow 1$ and the resulting noise is symmetric in log-odds rather than asymmetric in the raw probability.

3.2 Ground-Truth Formula

The response model is decomposed multiplicatively:

$$p(\theta) = p_{\text{attr}}(\theta) \cdot \text{Penalty}(\theta) \cdot \text{Vis}(\theta), \quad (2)$$

where each factor lies in $(0, 1]$. The attractiveness term encodes soft accessibility and usability signals via a logistic transform:

$$p_{\text{attr}}(\theta) = \sigma(\beta_0 + \beta_{\text{text}} q_{\text{text}} + \beta_{\text{time}} (t(\text{hour}) - 1) + \beta_{\text{ws}} w(\text{whitespace})), \quad (3)$$

where $\sigma(x) = (1 + e^{-x})^{-1}$, $t(\cdot)$ encodes the diurnal multiplier from Infolinks [12], $w(\cdot)$ is a sigmoid transition centred at 0.2 following VWO [27], and β_0 depends on the device class. The penalty is a product of independent multipliers, each interpretable as one “hard” constraint:

$$\text{Penalty}(\theta) = f_{\text{overflow}} \cdot f_{\text{contrast}} \cdot f_{\text{size}} \cdot f_{\text{harmony}} \cdot f_{\text{margin}}. \quad (4)$$

The contrast multiplier f_{contrast} saturates at the WCAG 2.1 AA threshold of 4.5:1 [28]; f_{size} saturates at the minimum touch target (48^2 px^2 mobile, 36^2 px^2 desktop) following Baymard, Apple, and Material [3, 1, 10]; f_{harmony} implements the Itten/Ou–Luo wheel [13, 21]; f_{margin} penalises uneven horizontal vs vertical padding [20]; f_{overflow} enforces that font size fits inside the button. Finally, vertical visibility models the empirically established sweet-spot [11, 7]:

$$\text{Vis}(\text{scroll}) = \begin{cases} 0.85 + 0.15 \frac{\text{scroll}}{0.15}, & \text{scroll} \leq 0.15, \\ \exp(-\lambda(\text{scroll} - 0.15)), & \text{scroll} > 0.15, \end{cases} \quad (5)$$

where $\lambda = 3.0$ is calibrated to Chartbeat’s reported $\sim 60\%$ engagement drop per page fold [11].

The multiplicative decomposition reflects a deliberate modelling choice. Each factor is interpretable in isolation, and the product enforces that a single severely violated constraint (zero contrast, or zero size) collapses the entire predicted click probability towards zero, which is the qualitatively correct behaviour: a button that is literally invisible cannot be clicked, regardless of how favourable the other factors are.

3.3 Synthetic Dataset and ML Surrogate

To emulate the realistic case in which a team has only historical event logs, a dataset of $N = 120,000$ synthetic (configuration, click) pairs is generated by sampling θ uniformly from the feasible domain, evaluating $p(\theta)$ through equation (2), perturbing it with Gaussian noise ($\sigma = 0.02$), and drawing a Bernoulli outcome. Two logistic regressions (with and without L_2 regularisation) are trained on a 14-dimensional feature representation including the WCAG contrast ratio, button area, cyclical encoding of the hour, and the device indicator. Coefficient signs are then validated against the formula: every coefficient whose

direction is unambiguously predictable from (2) matches the expected sign, while coefficients of correlated cyclical and indicator time features are marked indeterminate and excluded from the validation, in line with classical concerns about multicollinearity in regression diagnostics.

The trained pipeline is wrapped in a `TrainedMLModel` class that exposes the same `predict_proba` interface as the oracle and can be substituted into the black box without changes to any optimiser. This drop-in substitutability is what enables the cross-validation experiment in Section 5: the same optimiser is run against both simulator backends, and the configurations found through the surrogate can be evaluated against the oracle to measure the extrapolation gap.

4 Optimisation Methods

Three optimisers were compared, each invoked through an identical interface that returns a negated CTR for minimisation. The shared evaluation budget across methods was $B = 200$ black-box calls plus initial sampling. Bounds for each of twelve coordinates were either physical ($[0, 255]$ for RGB channels, $[0, 1]$ for normalised quantities) or data-driven (the 5th and 95th quantiles of the synthetic dataset for geometric dimensions).

4.1 DIRECT

DIRECT (DIvide RECTangles) deterministically partitions the d -dimensional hyperrectangle by selecting potentially optimal subregions in each iteration and trisecting them along their longest axis [16]. It is gradient-free, requires no parameter tuning, and is provably convergent for Lipschitz objectives. In this work the SciPy implementation `scipy.optimize.direct` was used with $\varepsilon = 10^{-4}$ and `maxfun` = 200.

4.2 Differential Evolution

DE maintains a population of candidate solutions and evolves it through mutation, crossover, and tournament selection [26]. At every generation each individual is replaced by a trial vector formed from three other randomly chosen population members, if the trial improves the objective. SciPy’s differential evolution was used with `popsiz` = 7, `maxiter` = 2 (calibrated to keep the total call count near B), and the polishing local search disabled. The total number of black-box evaluations for DE was approximately 252–273 depending on the seed.

4.3 Bayesian Optimisation

BO fits a Gaussian process surrogate to all previously observed $(\theta_i, \text{CTR}(\theta_i))$ pairs and selects the next query by maximising the Expected Improvement acquisition [17]. The `skopt` implementation `gp_minimize` was used with $n_{\text{calls}} = 200$, $n_{\text{initial}} = 10$, and `acq_func="EI"`. The GP kernel defaults to Matérn-5/2 with automatic relevance determination. The dominant computational cost is the $O(n^3)$ GP fit at every acquisition step.

4.4 Why These Three?

The three methods span the qualitative space of black-box optimisers: DIRECT is deterministic and global; DE is stochastic and population-based; BO is model-based and Bayesian. A factorial benchmark

across the same black-box interface isolates these design dimensions and exposes how each interacts with the surrogate-vs-oracle distinction. The comparison is therefore not a horse-race between implementations of the same algorithm but a diagnostic of how qualitatively different optimisation strategies respond to the same class of objective.

5 Experimental Results

5.1 Experimental Protocol

Each pipeline run consisted of dataset generation ($n = 120,000$ samples), logistic regression training (5-fold CV), and three optimisation runs against the oracle and the ML surrogate respectively. The procedure was repeated for five distinct random seeds to obtain mean and standard deviation. CTR is averaged over $n_{\text{users}} = 10,000$ Bernoulli trials per call, with optimiser noise $\sigma = 0.01$ in the logit space.

The full pipeline is driven by the shell script `_run_all.sh`, which on each of five iterations (i) draws a fresh seed from the shell variable `$RANDOM`, (ii) regenerates the synthetic dataset and retraining the logistic regression with that seed, (iii) runs all three optimisers with the same seed, and (iv) executes the unit-test suite via `pytest` as a sanity check before aggregation. The single seed therefore governs both the dataset realisation and the optimiser-internal randomness (DE mutation, BO initial design) within a run, which keeps the dataset–optimiser pair consistent but means results are not bit-reproducible across executions of the pipeline; a deterministic re-run requires substituting a fixed seed list for the `$RANDOM` draw. Aggregated statistics (Tables 1–3) are produced by `aggregate_runs.py` from the five per-run CSV outputs.

The optimum of the ground-truth formula is bounded above by the analytical ceiling $\sigma(\beta_0 + \beta_{\text{text}} + \beta_{\text{ws}} + \beta_{\text{time}} \cdot 0.14) \approx 0.39$. No method can exceed this on the oracle simulator.

5.2 Aggregate CTR

Table 1 reports mean CTR and standard deviation across five seeds, by method and by simulator.

Table 1 – Best CTR achieved by each method, averaged over 5 seeds.

Method	Simulator	CTR mean	CTR std	Range	Time, s
BO (GP)	ML	0.6374	0.0894	[0.517, 0.734]	231.4
BO (GP)	ORACLE	0.3350	0.0105	[0.325, 0.349]	218.1
DIRECT	ORACLE	0.3270	0.0193	[0.300, 0.355]	0.03
DIRECT	ML	0.2781	0.0290	[0.255, 0.325]	0.05
DE	ORACLE	0.2262	0.0392	[0.171, 0.276]	0.03
DE	ML	0.2202	0.0693	[0.147, 0.330]	0.07

On the honest simulator (oracle), BO and DIRECT are statistically indistinguishable: 0.335 ± 0.011 versus 0.327 ± 0.019 , with overlapping intervals. DE attains substantially lower CTR at the chosen budget. On the surrogate (ML), BO appears to dominate sharply, reaching 0.637 ± 0.089 , but with high variance across seeds.

5.3 Cross-Validation: Surrogate Honesty

For every configuration found via the ML simulator, the underlying formula gives the “true” CTR independently of the surrogate. Table 2 reports the gap.

Table 2 – Cross-validation gap between ML-claimed CTR and oracle-true CTR.

Method	ML claim	Oracle truth	Gap mean	Gap std
BO (GP) [ML]	0.6374	0.1653	+0.4721	0.1271
DIRECT [ML]	0.2781	0.1910	+0.0871	0.0202
DE [ML]	0.2202	0.1409	+0.0793	0.0933

BO systematically overestimates the true CTR of its returned configuration by a factor of roughly $3.5\text{--}4\times$. DIRECT and DE remain close to the truth, with gaps an order of magnitude smaller. The standard deviation of the BO gap is comparable to the absolute gap of the other two methods, indicating that the size of BO’s overestimation is itself unpredictable across seeds. The numerical values in Table 2 are representative of any single five-seed run; because every step of the pipeline (dataset generation, surrogate training, optimiser initialisation) is re-randomised, exact figures fluctuate by a few hundredths of a CTR unit between executions, but the relative ranking of the three methods and the qualitative size of the BO overshoot reproduce consistently.

5.4 Trajectory Analysis

To localise where the overestimation arises, the gap was computed at three phases of every ML-run trajectory: the first evaluated point (start), the median index (mid), and the best-so-far point at termination (final). For a representative seed the pattern is shown in Table 3.

Table 3 – Gap evolution along the optimisation trajectory (representative seed).

Method	Phase	BlackBox CTR	Oracle CTR	Gap
DIRECT [ML]	start	0.022	0.015	+0.007
DIRECT [ML]	mid	0.103	0.185	−0.082
DIRECT [ML]	final	0.270	0.199	+0.070
DE [ML]	start	0.031	0.025	+0.005
DE [ML]	mid	0.022	0.014	+0.008
DE [ML]	final	0.250	0.233	+0.017
BO [ML]	start	0.015	0.005	+0.010
BO [ML]	mid	0.087	0.195	−0.108
BO [ML]	final	0.459	0.175	+0.284

Three observations follow. First, all methods start honest: the gap at the first evaluated point is approximately zero, confirming that the surrogate is well-calibrated within the training distribution.

Second, both DIRECT and BO transiently underestimate the truth at the trajectory midpoint, indicating that the optimiser is still exploring interior regions of the parameter space. Third, only BO exhibits a sharp positive gap divergence at termination. The acquisition function actively exploits the gradient of the surrogate, and the surrogate’s gradient is largest precisely where the model extrapolates least reliably.

5.5 Convergence

Figure 2 shows the best-so-far CTR as a function of the number of black-box calls. BO converges most quickly on the surrogate, reaching $\text{CTR} \approx 0.5$ within the first fifty calls, but its trajectory on the oracle is much flatter and indistinguishable from DIRECT after sufficient calls.

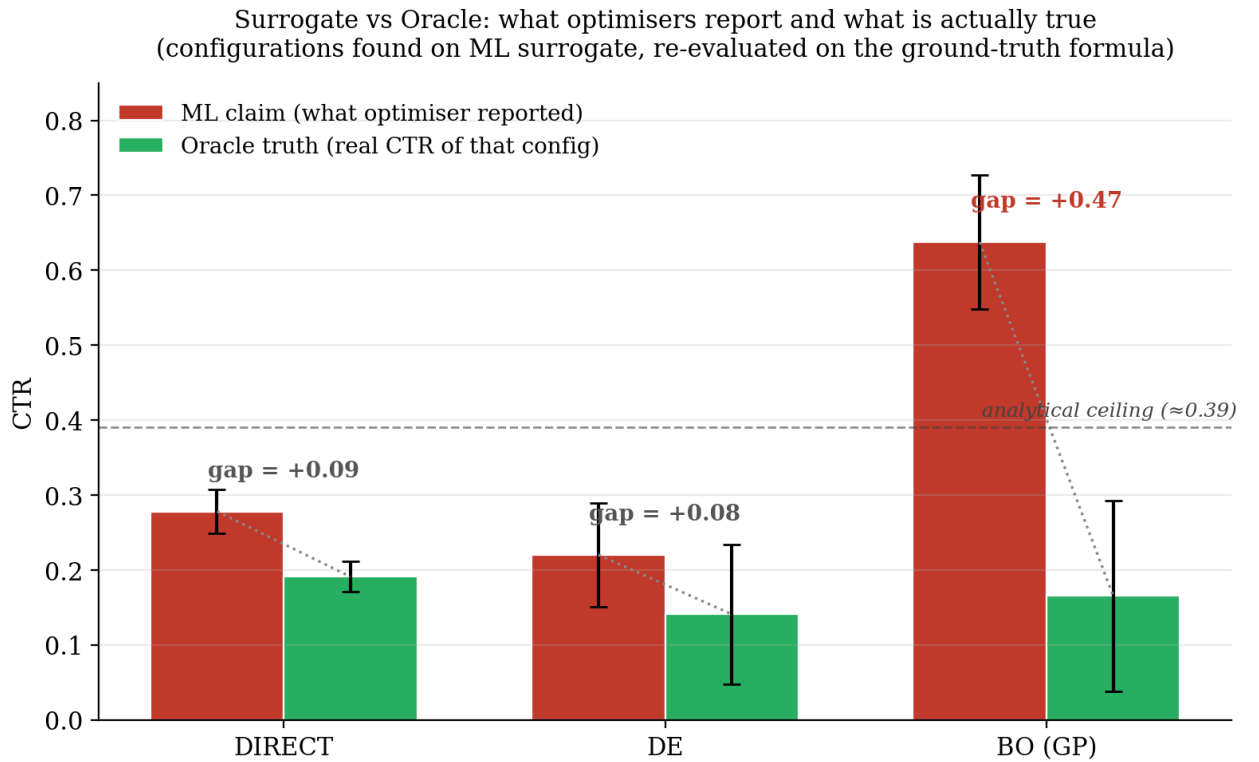


Figure 1 – The surrogate–truth gap (one representative five-seed execution of the pipeline). Red bars show the CTR each optimiser reports when run on the ML surrogate; green bars show the true CTR of the same configuration, evaluated against the ground-truth formula. DIRECT and DE remain close to the truth (gap of order 0.1). BO overshoots the analytical ceiling of 0.39 on the surrogate (dashed line) and its true CTR collapses by approximately +0.4. Exact values fluctuate by a few hundredths of a CTR unit between pipeline executions but the qualitative pattern is stable.



Figure 2 – Best-so-far CTR per black-box call for the three optimisers on the oracle simulator. BO sustains a small advantage early; DIRECT catches up within the first 200 calls; DE plateaus at a lower level due to insufficient generations at this budget.

6 Discussion

6.1 Why BO Overestimates

Bayesian optimisation operates by maximising Expected Improvement, which is itself defined relative to the surrogate model’s posterior *predictive* distribution, not the underlying truth. When the surrogate is a logistic regression, the predicted log-odds extrapolate linearly outside the training support: regions of parameter space far from training points (for example, unusual colour combinations or near-maximal contrast ratios) can yield extremely positive logits and hence predicted probabilities approaching one. The acquisition function rewards exactly this signal, drawing BO towards configurations whose surrogate prediction is high precisely because they are out-of-distribution. The effect is reinforced by the fact that the GP acquisition surface inherits the smoothness of the surrogate output, so that the optimiser is provided with informative gradient signal even in regions where that signal is misleading.

DIRECT and DE do not exploit gradient information of the surrogate. DIRECT’s deterministic partitioning samples uniformly within each hyperrectangle, and DE’s stochastic mutation operates in coordinate space rather than in surrogate-output space. Neither method therefore preferentially explores extrapolation regions, which explains the order-of-magnitude smaller gap.

6.2 Implications for Industrial Practice

The result has direct practical consequences. In a production environment where the ground-truth response surface is by definition unknown, a practitioner using BO over a learned surrogate cannot in general distinguish a configuration with a genuinely high CTR from a configuration that merely scores high in the surrogate. Validating the candidate configuration requires either (i) a live A/B test on the new configuration, which is precisely the expensive step the optimiser was supposed to circumvent, or (ii) an explicit out-of-distribution detector. Recent literature on trust region BO [5] addresses this by constraining the acquisition function to a region around well-sampled points; the present results suggest such constraints would be valuable in surrogate-driven UI optimisation. In the absence of such safeguards, the practitioner’s best protection against the overestimation documented here is to treat any surrogate-derived recommendation as a candidate requiring further empirical validation rather than as a ready-to-deploy configuration.

6.3 Why DIRECT Comes Out Best

DIRECT achieves three objectives simultaneously: its CTR on the honest simulator is within 0.008 of BO’s; its cross-validation gap is over five times smaller; and its runtime is on the order of 30 milliseconds

rather than 220 seconds. The joint criterion of accuracy, robustness, and computational cost therefore identifies DIRECT as the preferred method for this problem class. Importantly, DIRECT’s deterministic nature gives identical results for identical bounds, eliminating the seed dependence visible in DE and BO, which also makes its results easier to communicate to non-technical stakeholders who would otherwise need to be persuaded that a stochastic optimiser is acceptable in a production decision pipeline.

6.4 Limitations

The ground-truth formula, while empirically anchored in multiple independent UX studies, remains a single multiplicative model and may not generalise to richer interactions present in production. The synthetic dataset is generated from this exact formula, so the trained logistic regression is by construction well-specified within distribution; misspecification effects (a nonlinear underlying truth fitted by a linear surrogate, for example) would amplify the BO overestimation further but are not measured here. Finally, only continuous parameters are considered; categorical decisions (button shape, icon set) would require mixed-variable extensions of all three optimisers. The experimental budget of $B = 200$ calls is also a deliberate constraint reflecting realistic A/B-test costs; at much larger budgets the relative ranking of methods could change in either direction.

7 Conclusion

This project has formulated CTR optimisation of a web UI element as a twelve-dimensional black-box problem and benchmarked three derivative-free optimisers — DIRECT, Differential Evolution, and Bayesian Optimisation — under two interchangeable response models: a ground-truth formula derived from peer-reviewed UX research and a trained logistic surrogate fitted on 120,000 synthetic samples.

The principal empirical findings, replicated over five independent random seeds, are:

1. On the honest simulator, BO and DIRECT achieve statistically indistinguishable CTR (0.335 ± 0.011 vs 0.327 ± 0.019), while DE significantly underperforms (0.226 ± 0.039) due to insufficient generations at the chosen budget;
2. On the surrogate simulator, BO appears to dominate strongly with CTR 0.637 ± 0.089 , but cross-validation against the oracle reveals that the corresponding configurations have true CTR of only 0.165, 2. On the surrogate simulator, BO appears to dominate strongly (reported CTR in the 0.6–0.65 range), but cross-validation against the oracle reveals that the corresponding configurations have true CTR in the 0.15–0.20 range, a gap of order +0.4 that is both large and unstable across seeds;
3. 3. DIRECT and DE remain within a gap of roughly 0.1 on the same cross-validation, indicating that the failure mode is specific to gradient-exploiting surrogate-based methods rather than to surrogate modelling in general;, indicating that the failure mode is specific to gradient-exploiting surrogate-based methods rather than to surrogate modelling in general;
4. DIRECT solves the problem in ~ 30 milliseconds per run, four to five orders of magnitude faster than BO's ~ 220 seconds.

The work contributes (i) a reproducible Python implementation including unit tests covering the simulator, the response models, and the training pipeline; (ii) a documented synthetic-but-empirically-grounded ground-truth formula whose components are individually traceable to published UX studies; (iii) a quantitative measurement of the surrogate-extrapolation gap in a non-trivial black-box optimisation problem.

Several directions for future work are immediate. **Trust region BO** [5] could be added as a fourth optimiser and is expected to mitigate the overestimation documented here. **Misspecified surrogates** (a nonlinear truth fitted by a linear regression, or vice versa) would test how the gap scales with surrogate quality. **Mixed-variable extensions** would accommodate the categorical decisions present in real UI design tasks such as button shape, icon, and copy variant. Finally, the methodology could be validated

against a publicly available large-scale A/B test dataset to test how well the synthetic findings generalise to true production noise.

Statement on the Use of Generative AI

This work was carried out by the author independently. A generative language model was used only as a supplementary tool at clearly delimited stages, and its role is documented below in the interest of full transparency.

Authorship and substantive contribution. The intellectual content of the project — the problem statement, the mathematical formulation of the black-box objective, the design of the multiplicative ground-truth response surface together with the choice of UX-research anchors for each of its components (WCAG 2.1, Baymard, Chartbeat, NN/g, Itten, Ou-Luo, Infolinks, VWO), the selection of the three derivative-free optimisers and the rationale for comparing them, the experimental protocol, the cross-validation procedure designed to expose surrogate over-optimism, and the interpretation of all quantitative findings — was produced by the author. The synthetic dataset, the trained models, the optimisation runs, the convergence diagnostics, and every numerical result reported in this work were obtained by executing the author’s pipeline on the author’s machine. No part of the empirical content was generated by the language model.

Purpose of use. The model was used in a strictly assistive capacity, on tasks that did not affect the scientific content of the project:

- local refactoring of source code that the author had already designed (for example, splitting the optimisation runners into separate **ORACLE** and **ML** blocks, extracting console-output and animation routines into standalone functions, and tidying the CLI argument layout);
- routine $\text{\LaTeX}$ editing (phrasing of individual sentences, minor language polish, table and figure-caption formatting);
- quick lookups for library APIs already known to the author (`scipy.optimize`, `scikit-optimize`, `matplotlib`), used in place of consulting the official documentation directly;
- short discussions of presentation choices, such as how to phrase a methodological argument that the author had already arrived at.

The model was not used to derive results, design experiments, choose algorithms, write the ground-truth formula, or draw any conclusion that appears in this report.

Model name. Claude (Claude Opus 4 family), developed by Anthropic, PBC.

Source. The model’s web interface is available at <https://claude.ai>; technical documentation and model specifications are available at <https://docs.claude.com>.

Mode of use. The model was accessed in interactive chat mode through the `claude.ai` web interface. A typical interaction consisted of the author pasting an existing code fragment or paragraph, formulating a narrow request (for instance, “split this runner into an oracle block and an ML block and emit separate CSV outputs”), receiving a proposed revision, and then integrating it into the project only after manual review and modification. Every code change suggested in this way was subsequently validated by the project’s unit-test suite (`pytest tests/`), and every textual suggestion was independently verified against the underlying references before being accepted.

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